PHYSIOLOGY

Cardiac Physiology

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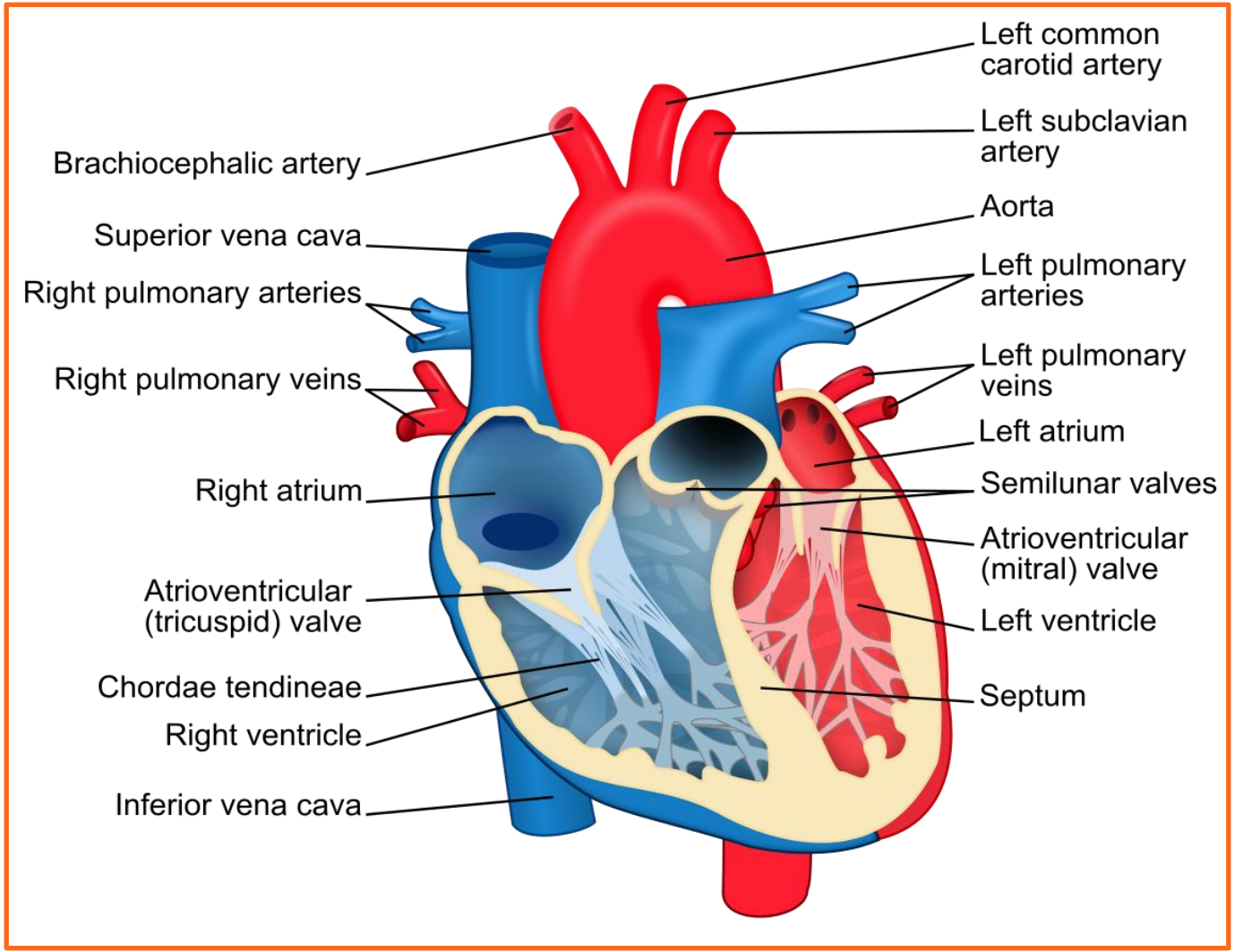
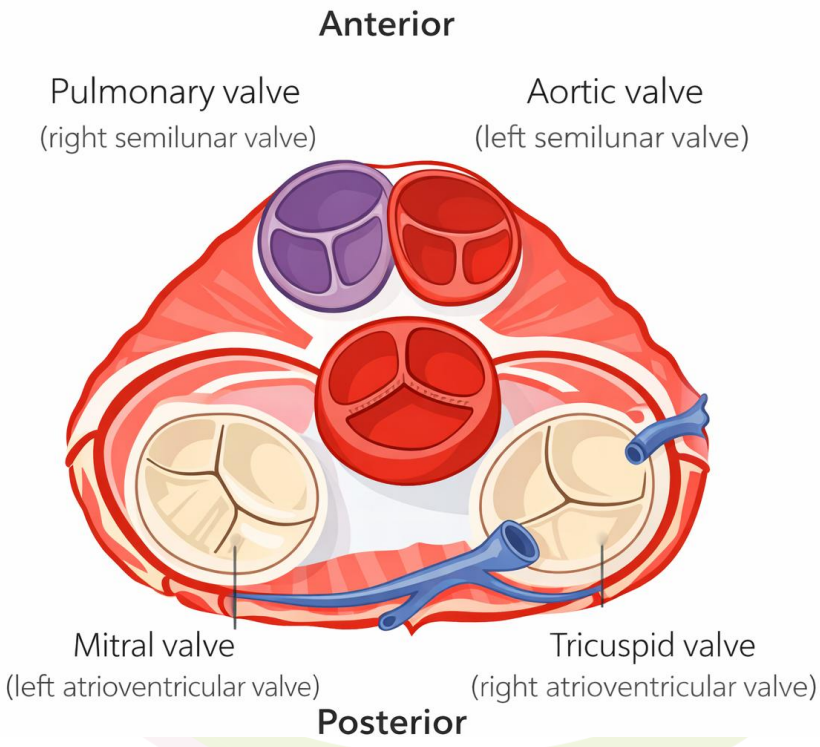
Let's Learn Logically


Left common carotid artery
Left subclavian artery
Aorta
Left pulmonary arteries
Left pulmonary veins
Left atrium
Semilunar valves
Atrioventricular (mitral) valve
Left ventricle
Septum
Brachiocephalic artery
Superior vena cava
Right pulmonary arteries
Right pulmonary veins
Right atrium
Atrioventricular (tricuspid) valve
Chordae tendineae
Right ventricle
Inferior vena cava

Heart Valves

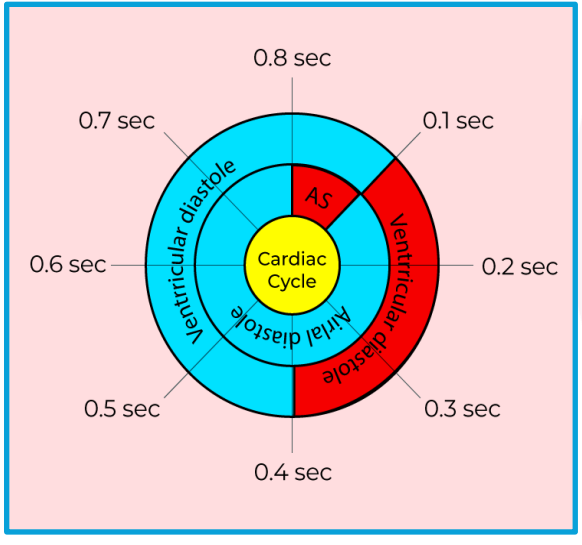


Cardiac Physiology

- Heart is the vital organ.
- Pumps blood.
- Made of four Chambers – **Two atria & Two ventricles**

Properties of heart -

S.no	Properties	Meaning
1	**Excitability**	Ability to respond to stimulus
2	**Rhythmicity**	Ability to produce its own impulses regularly
3	**Conductivity**	Conduction of impulse through heart
4	**Contractility**	Ability of tissue to shorten in length



Cardiac cycle - 0.8 sec

Atrial events

Ventricular events

Atrial systole 0.11 sec

Atrial diastole 0.69 sec

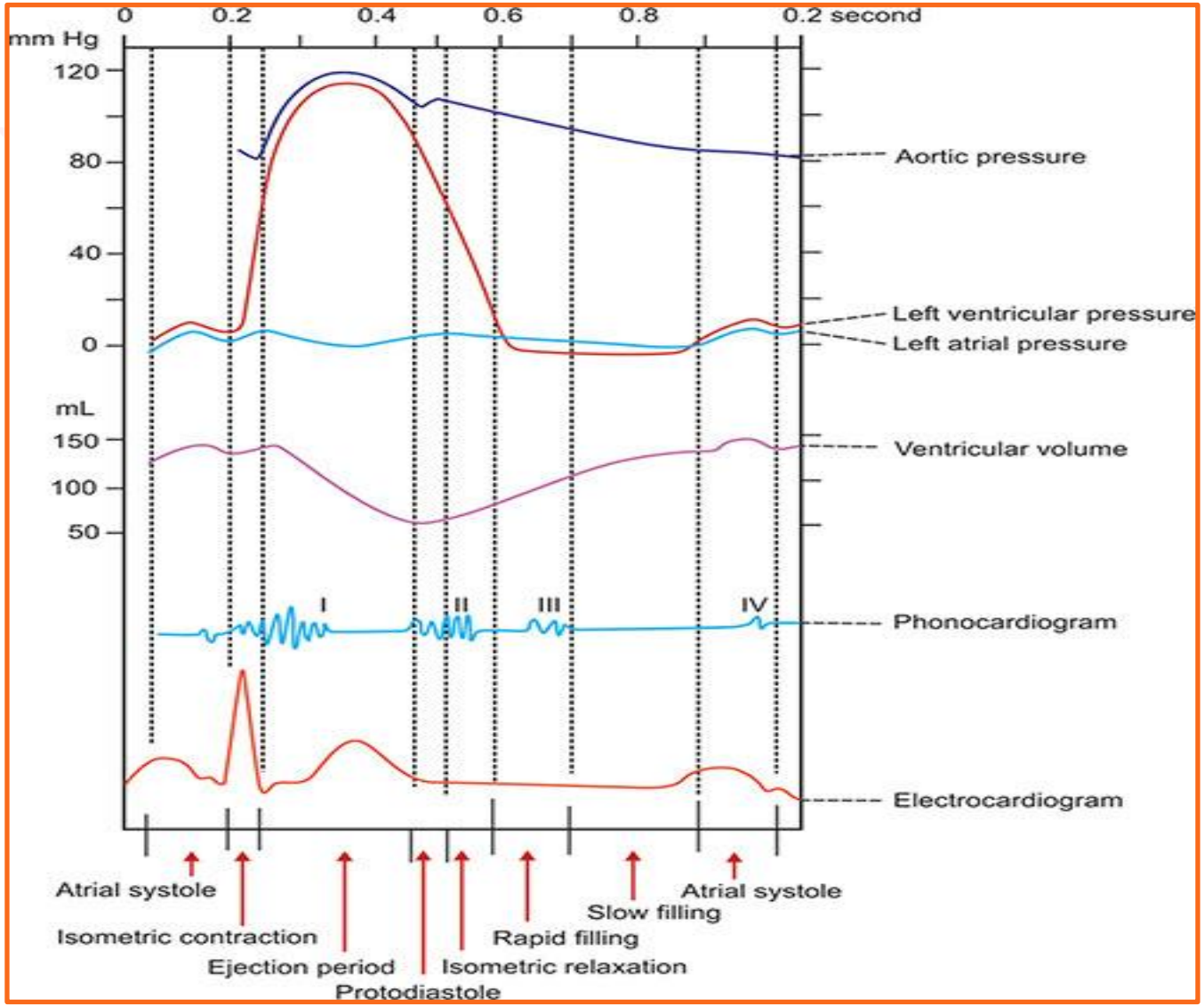
Ventricular systole 0.30 sec

Ventricular diastole 0.53 sec

1. Isovolumetric contraction - 0.05 sec
2. Ejection phase - 0.25 sec

Rapid ejection - 0.10 sec
Reduced ejection - 0.15 sec

Proto-diastolic-phase- 0.04sec
Isovolumetric relaxation - 0.06 sec
Filling phase - 0.4 sec
First rapid filling phase - 0.10 sec
Diastasis - 0. 20 sec
Last rapid filling phase - 0.10 sec


0
0.2
0.4
0.6
0.8
0.2 second
mm Hg
120
80
40
0
mL
150
100
50
Aortic pressure
Left ventricular pressure
Left atrial pressure
Ventricular volume
I
II
III
IV
Phonocardiogram
Electrocardiogram
Atrial systole
Isometric contraction
Ejection period
Protodiastole
Isometric relaxation
Rapid filling
Slow filling
Atrial systole

Frank Starling Law

- Within physiological limits, Force of contraction is directly proportional to Initial length of the fibre.
- Here, Force of contraction is directly proportional to End diastolic fibre length.
- **Pre-load –** The load which operates before the beginning of contraction. (End diastolic volume)
- **After- load –** Begins to operate after the onset of the ventricular contraction. (Aortic Pressure)
- Pace-maker of heart= SA node (Site- At the junction of Sup. Vena cava with right atrium)

Tissues	Rate (beat/minute)	Conduction speed (m/sec)
SA node	70-80	0.05
AV node	40-60	0.05
AV bundle	30-36	1
Purkinje fibers	15-40	4

Heart sounds		
Features	**First heart sound**	**Second heart sound**
Cause	Closure of AV valve	Closure of semilunar valve
Characteristics	Long, soft, low pitched	Short, sharp, high pitched
Occurs during	Isometric contraction	Protodiastole
Relation with ECG	**Coincides with R wave**	**After T wave**

Cardiac murmur		
Systolic murmur	**Diastolic murmur**	**Continuous murmur**
‣ Incompetence of AV valves ‣ Stenosis of semilunar valves ‣ Anemia ‣ Septal defect ‣ Coarctation of aorta	‣ Stenosis of AV valves ‣ Incompetence of semilunar valves	‣ Patent ductus arteriosus

ECG

- Electrocardiogram (ECG) is the graphical presentation of the electrical activities of heart.
- Speed of ECG paper – 25 mm/sec
- ECG machine records the difference of potential between two points.

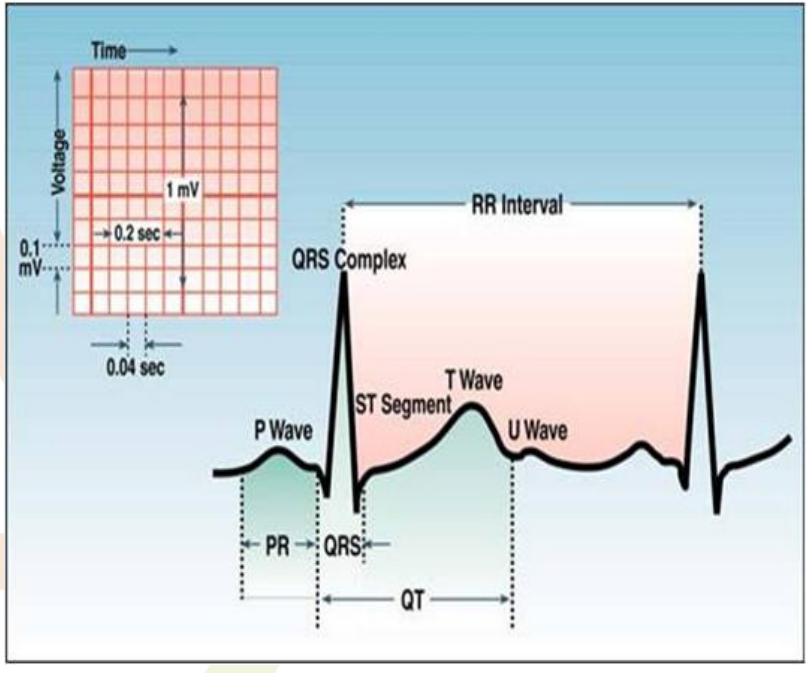
- **Einthoven triangle –** An equilateral triangle that is used as a model of standard limb leads used to record ECG. Heart is supposed to be the centre.

ECG leads

- Bipolar leads - Limb lead I, Limb lead II, Limb lead III
- Unipolar leads

  Unipolar limb leads - aVR lead, aVL lead, aVF lead

  Unipolar chest leads - V1, V2, V3, V4, V5, V6



- P wave – Atrial depolarization (0.08sec – 0.12 sec)
- QRS complex – Ventricular depolarization (0.08 – 0.12 sec)
- T wave – Ventricular repolarization (0.27 sec)

HEART RATE
(NORMAL ECG)

HEART RATE = 300 ÷ (NUMBER OF LARGE SQUARES IN ONE R–R INTERVAL)

300 ÷ 4 = 75 BPM

HEART RATE = 75 BPM

Lead I - left forearm and right forearm

Lead II - right forearm and left leg

Lead III - left leg and left forearm

V1 = 4th intercostal space , right border of the sternum

V2 = 4th intercostal space, lift border of the sternum

V3 = midway between v2 and v4

V4 = 5th intercostals space at mid clavicular line

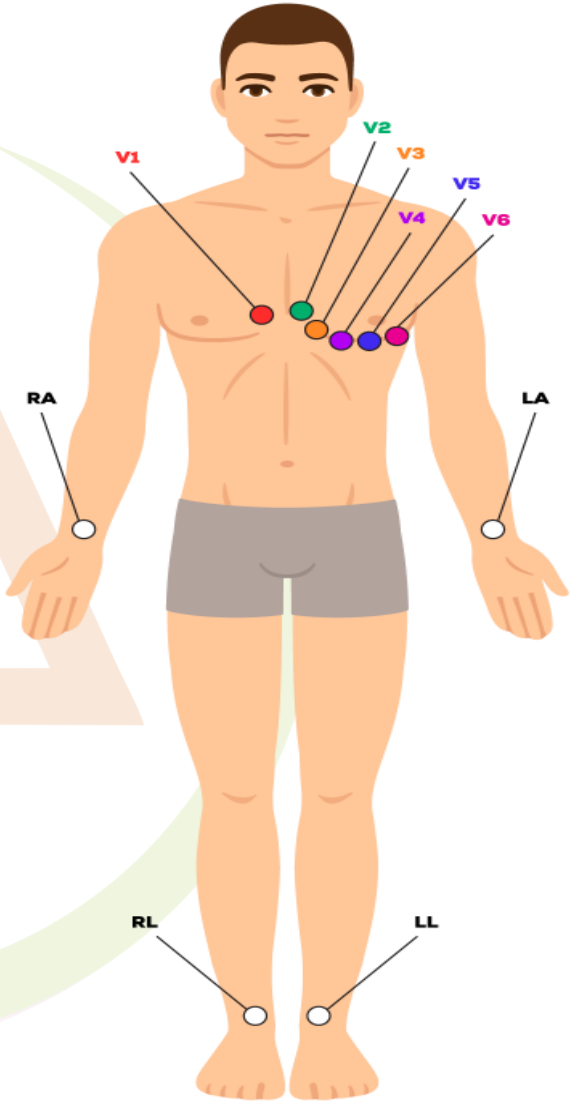
V5 = 5th intercostals space at ant. Axillary line

V6 = 5th intercostals space at mid axillary line

aVR = right forearm

aVR = left forearm

aVF = left foot



- **Stroke volume** – Amount of blood ejected out by each ventricle per beat = 70 ml
- **Cardiac output** – Volume of blood ejected by each ventricle per minute

CO = SV x HR (70 X 70 = 4900 ml or 4.9 L)

- **Ejection fraction** – Fraction of end- diastolic volume that is ejected out by each ventricle. 60 – 65%
- Liver receives maximum percentage of blood i.e. 30%.

- **Heart rate** – 60 - 100 per minute (72 beat/ minute)
  - **Tachycardia** – Increase in heart rate
  - **Bradycardia** – Decrease in heart rate

Blood pressure – Lateral pressure exerted by the moving column of blood on the walls of vessels.

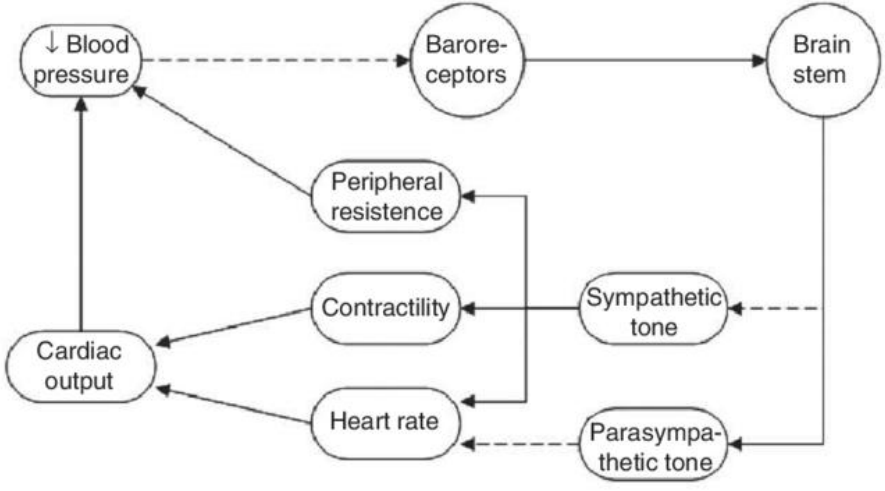
BP = CO x PR

BP sounds – Korotkoff sounds

- Tapping (Systolic BP)
- Swissing
- Banging
- Muffled (Disappearance- Diastolic BP)

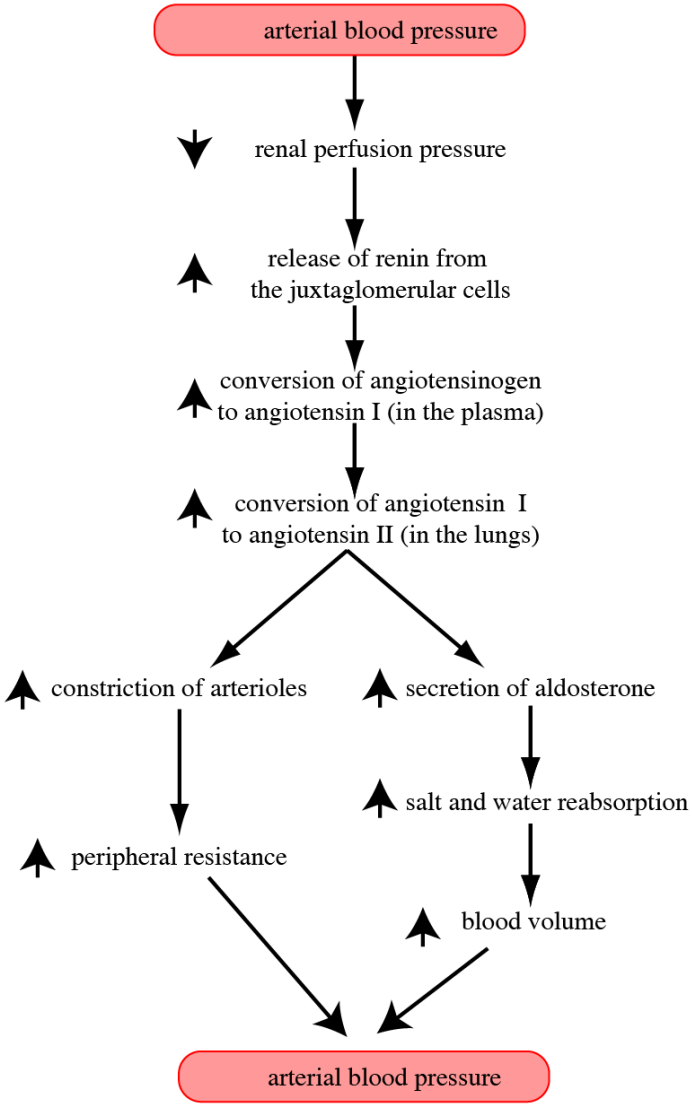
Regulation of Blood Pressure

Short term regulation (Carotid sinus & Aortic Arch)



Regulation of Blood Pressure

Long term regulation



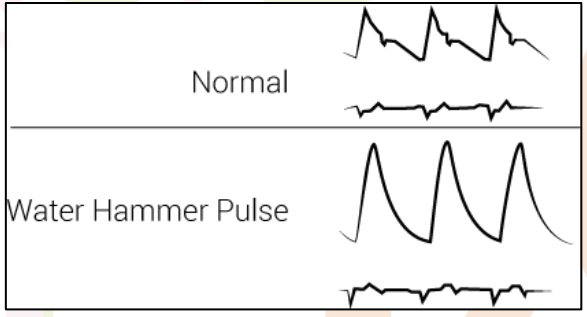
- Systolic Blood pressure – 110 - 140 mm Hg (120 mm Hg)
- Diastolic Blood pressure – 60 - 80 mm Hg (80 mm Hg)
- Pulse pressure = Systolic BP – Diastolic BP (120 – 80 = 40 mm Hg)
- Mean Arterial pressure = Diastolic Pressure + 1/3 of Pulse Pressure

  (80 + 40/3 = 93.3 mm Hg)

Clinical Examination of Pulse

1) Rate
2) Rhythm
3) Volume
4) Force
5) Equality on both sides
6) Condition of vessel wall

Water Hammer Pulse- Sudden strong bounding pulse which collapse rapidly. Eg. Aortic Regurgitation, High stroke volume



Pulse Deficit- Difference in Heart rate and Pulse rate (Missing pulse) (Eg. Atrial Fibrilation >10/min)

Weak/thready pulse- A small pulse that feels like a wire or thread palpation of arteries.

Paradoxical pulse/ Pulsus paradoxus -Force of strength of pulse wave varies, feeling weaker at time of inspiration. If <10 mmHg (Normal); If >10mmHg (Clinical sign). Cardiac causes- Cardiac tamponade, Constructive Pericarditis. Respiratory causes- Severe asthma, Tension pneumothorax